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PREAMBLE

DO NOT REMOVE THE LINE BELOW

```
clear; close all;
```

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```

1. COMMENTING

```
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```

```
type('eel3135_lab01_comment.m')
```

```
% Copy and comment every line of the following MATLAB script. Say what
% each line is doing in your comment. Explain each MATLAB line by using
% no more than one comment line, as done in the first line below. Run and
% publish the script:
a=zeros(1,10) % Generate and print a 1x10 row vector of zeros
b=ones(4,2) %Create and print a 4x2 matrix of ones, store in b
c=size(b); %stores the rows and columns of the b matrix in c (4,2)
abs([-2.2 , 3]) %returns the absolute values of the elements in the vector
floor(1.6) %rounds the value down to the closest integer
d=[1:-2.5:-9]; %create a row vector starting at 1, decrementing by 2.5,
and ending at -9
f=d(2); %assigns the second element of d to f
g=sin(pi/2); %computes sine of pi/2 and stores in g
```

```

K=[1.4, 2.3; 5.1, 7.8]; % defines a 2x2 matrix with these values
m=K(1,2); %sets m to the element in the first row and second column of
K
n=K(:,2); %sets n to the second column of K
p=K(1,2); %sets p to the element in the first row and second column of K
(same as m)
comp = 10+40i; %creates a complex number
real(comp) %returns the real part of the complex number
imag(comp) %returns the imaginary part of the complex
number
abs(comp) %returns the magnitude of the complex number
angle(comp) %returns the phase angle of the complex number in
radians
disp('haha, MATLAB is fun'); %prints the text in the command window
3^2 %computes 3 to the second power
4==4 %evaluates if 4 is equal to 4
[2==8 3~=5] %perform logical comparisons and store the results in a
vector
x=[0:3:9]; %creates array starting at 0 incrementing by 3 and going to 9,
store in x
y=[5 9 6 8]; %creates array with values 5,9,6,8 store in y
tic; pause(0.2); toc % Measure elapsed time with a pause of 0.2
seconds

q = zeros(10,1); %creates 10x1 column vector of zeros
for ii = 1:10 % Begin a loop that runs from 1 to 10
    q(ii) = ii^2; % Store the square of the loop index in q
end
figure(129); % Open a figure window with figure number 129
stem(x,y) % Create a stem plot of y versus x
hold on; % Retain the current plot when adding new plots
plot(x,y, 'k', 'linewidth', 2) % Plot y versus x as a black line
plot(x,y,'+r', 'markersize', 20); % Plot red plus markers at the data points
hold off; % Release the plot hold
xlabel('Horizontal Axis') % Label the x-axis
ylabel('Vertical Axis') % Label the y-axis
snapnow; % Capture the current figure for publishing

```

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=====

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2. PLOTTING SINUSOIDS

```

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=====

```

```

% -----
% 2 (a)
% -----

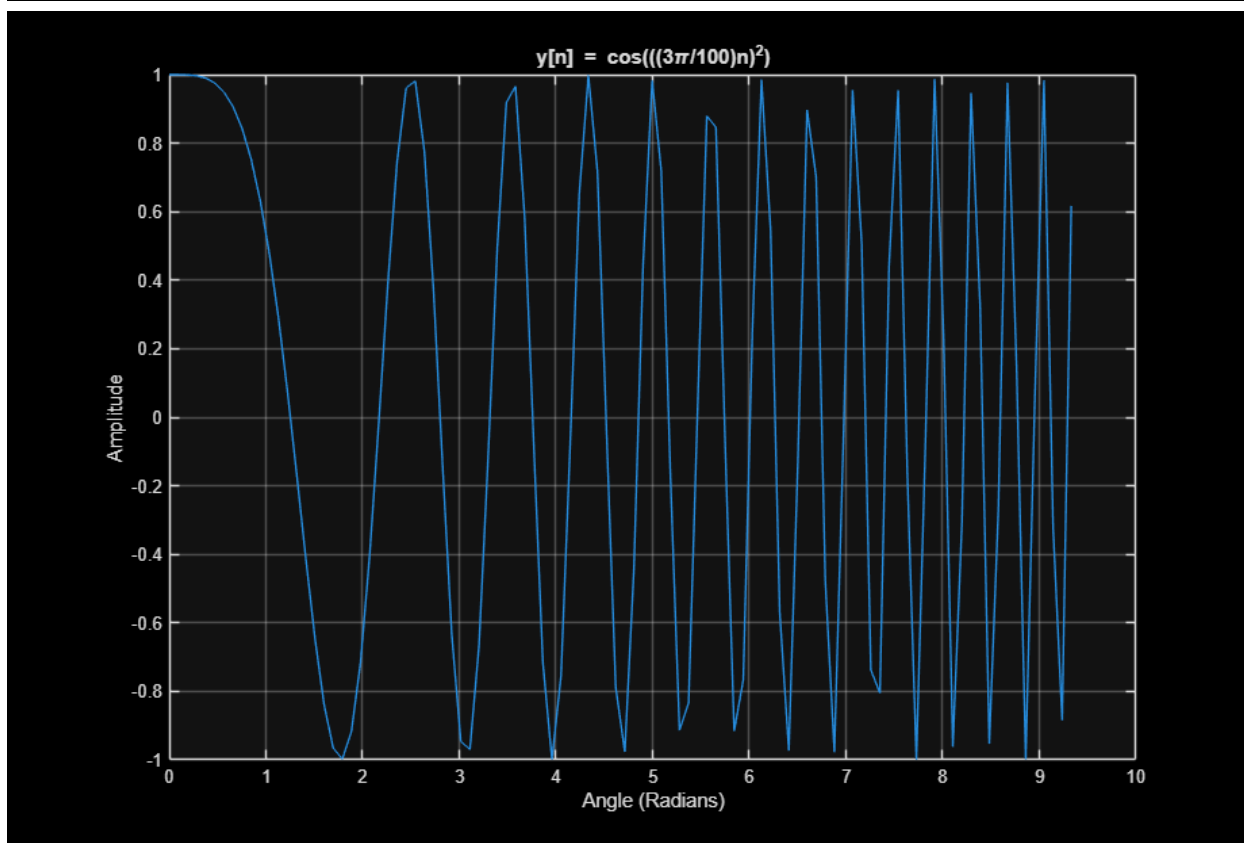
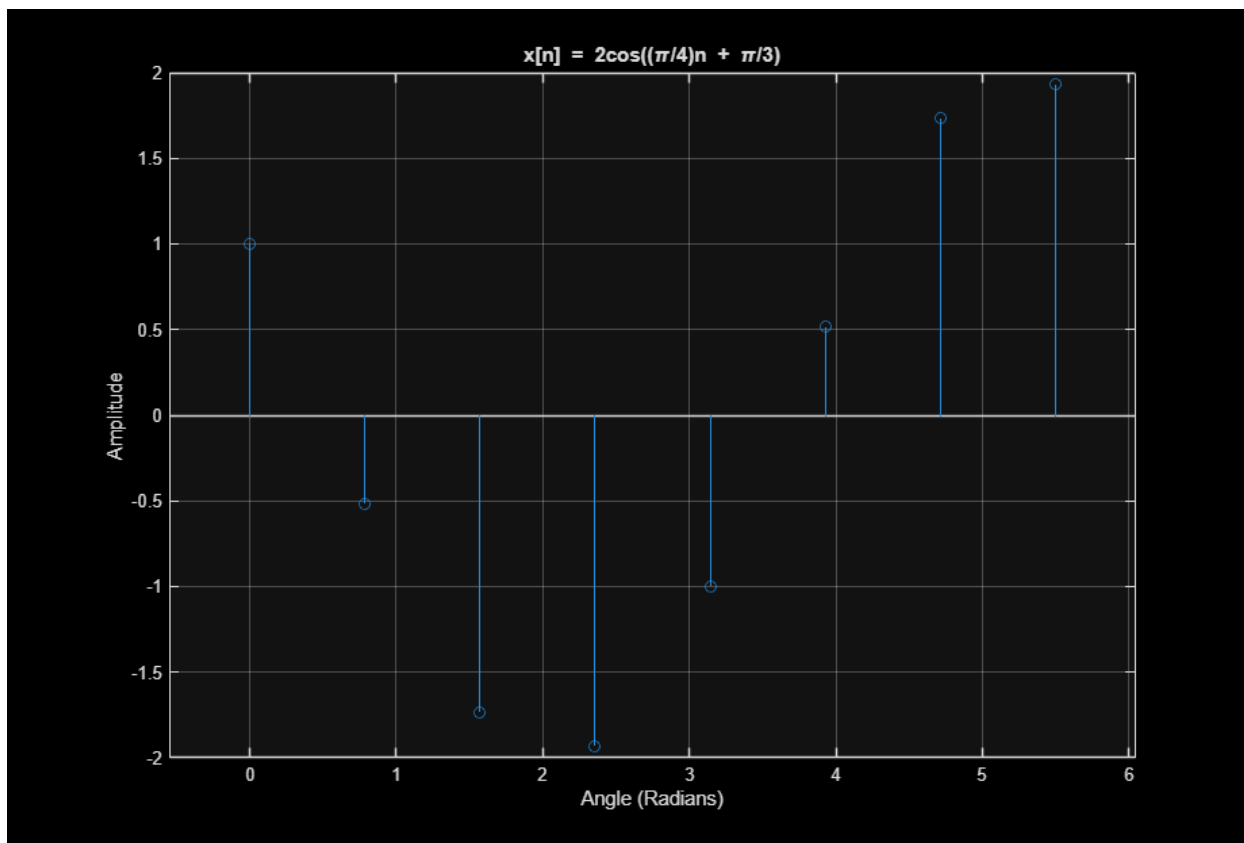
```

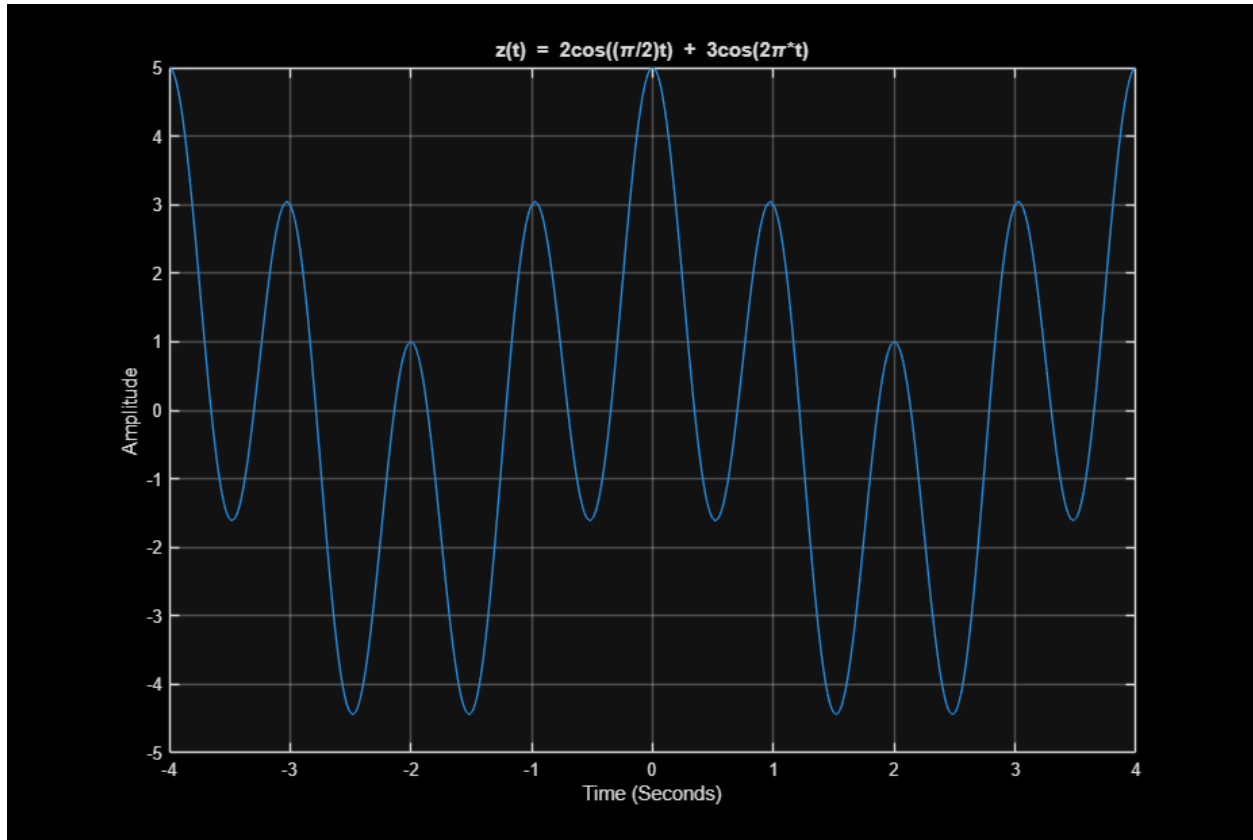
```

n= 0:7; %define sample indices from 0 to 7
theta = (pi/4)*n; %x-axis increases by pi/4 at a time
x = 2*cos((pi/4)*n + pi/3); %x[n] = 2cos(( $\pi/4$ )n +  $\pi/3$ )
figure
stem(theta, x); %plot function at theta values
xlabel('Angle (Radians)');
ylabel('Amplitude');
title('x[n] = 2cos(( $\pi/4$ )n +  $\pi/3$ )'); %added title and grid
grid on;
% -----
% 2(b)
% -----
n= 0:99; %define sample indices from 0 to 99
theta = (3*pi/100)*n; %x-axis increases by 3pi/100 at a time
y = cos(theta.^2); %y is cos function y[n] = cos[(3 $\pi/100$ )n]^2
figure
plot(theta, y); %plot function at theta values
xlabel('Angle (Radians)');
ylabel('Amplitude');
title('y[n] = cos((3 $\pi/100$ )n)^2'); %added title and grid
grid on;

% -----
% 2(c)
% -----
t = -4:0.01:4; %define t vector samples
z = 2*cos((pi/2)*t) + 3*cos(2*pi*t); %z function
figure
plot(t, z); %plot function at t values
xlabel('Time (Seconds)');
ylabel('Amplitude');
title('z(t) = 2cos(( $\pi/2$ )t) + 3cos(2 $\pi$ *t)'); %added title and grid
grid on;

```





```
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=====
```

3. FINDING COMPLEX ROOTS

```
=====
=====
```

```
% -----
% 3 (a)
% -----
type('myroots.m')

% -----
% 3 (b)
% -----
% ANSWER QUESTION BELOW
% The purpose of the help command is to display the documentation comments
% at the top of a MATLAB function file. It allows users to see what the
% function does and what the inputs and outputs should be without having to
```

```

% open the code.
%

% -----
% 3(c)
% -----
disp(myroots(5, -1));
disp(myroots(11, 2048j));

function r = myroots(n, a)
% myroots: Find all the nth roots of the complex number a
%
% Input Args:
%   n: a positive integer specifying the nth roots
%   a: a complex number whose nth roots are to be returned
%
% Output:
%   r: 1xn vector containing all the nth roots of a

r = zeros(1, n); % create a 1xn vector filled with 0

if a == 0 %handle 0 case
    r(:) = 0; %all nth roots of 0 are 0
    return
end

A = abs(a); % Compute magnitude A of complex number a
phi = angle(a); % Compute phase angle of complex number a
rootMag = A^(1/n); % Compute the magnitude of each nth root

for k = 0:n-1 % Loop over k to generate all n distinct roots
    r(k+1) = rootMag * exp(1j*(phi + 2*pi*k)/n); % Compute kth root using
    polar form formula
end

end

Columns 1 through 4

    0.8090 + 0.5878i   -0.3090 + 0.9511i   -1.0000 + 0.0000i   -0.3090 - 0.9511i

Column 5

    0.8090 - 0.5878i

Columns 1 through 4

    1.9796 + 0.2846i    1.5115 + 1.3097i    0.5635 + 1.9190i   -0.5635 + 1.9190i

Columns 5 through 8

```

$-1.5115 + 1.3097i$ $-1.9796 + 0.2846i$ $-1.8193 - 0.8308i$ $-1.0813 - 1.6825i$

Columns 9 through 11

$-0.0000 - 2.0000i$ $1.0813 - 1.6825i$ $1.8193 - 0.8308i$

4. AI FOR SIGNALS

-----4(a)-----

%You are acting as a MATLAB signal-processing assistant. Generate a MATLAB script to produce a signal $x(t)$ defined on $t = 0$ to $t = 2$ seconds, sampled at $f_s = 200$ samples per second.

%The signal must satisfy all of the following properties:

- %• Exactly 3 distinct nonzero frequency components.
- %• The signal must be exactly periodic.
- %• The signal must have a mean of 1.

%The output must:

- %1. Define the time vector t
- %2. Define $x(t)$ using sinusoidal components
- %3. Plot $x(t)$ with proper axis labels and title

% -----
% 4 (b)
% -----

% -----
% MATLAB script - Periodic sine-series with mean = 1
% -----

% Sampling parameters

$f_s = 200;$ % Samples per second
 $T = 2;$ % Total duration (seconds)

% Time vector (avoid repeating the end point so that the signal
% is exactly periodic over the interval)

$t = 0 : 1/f_s : T-1/f_s;$ % 0 ... 1.99 s

% ----- Frequency components -----

% Fundamental period = 2 s $\rightarrow$ fundamental frequency $f_0 = 1/T = 0.5$ Hz

```

f = [0.5    1.0    1.5];    % three distinct, non-zero frequencies (Hz)

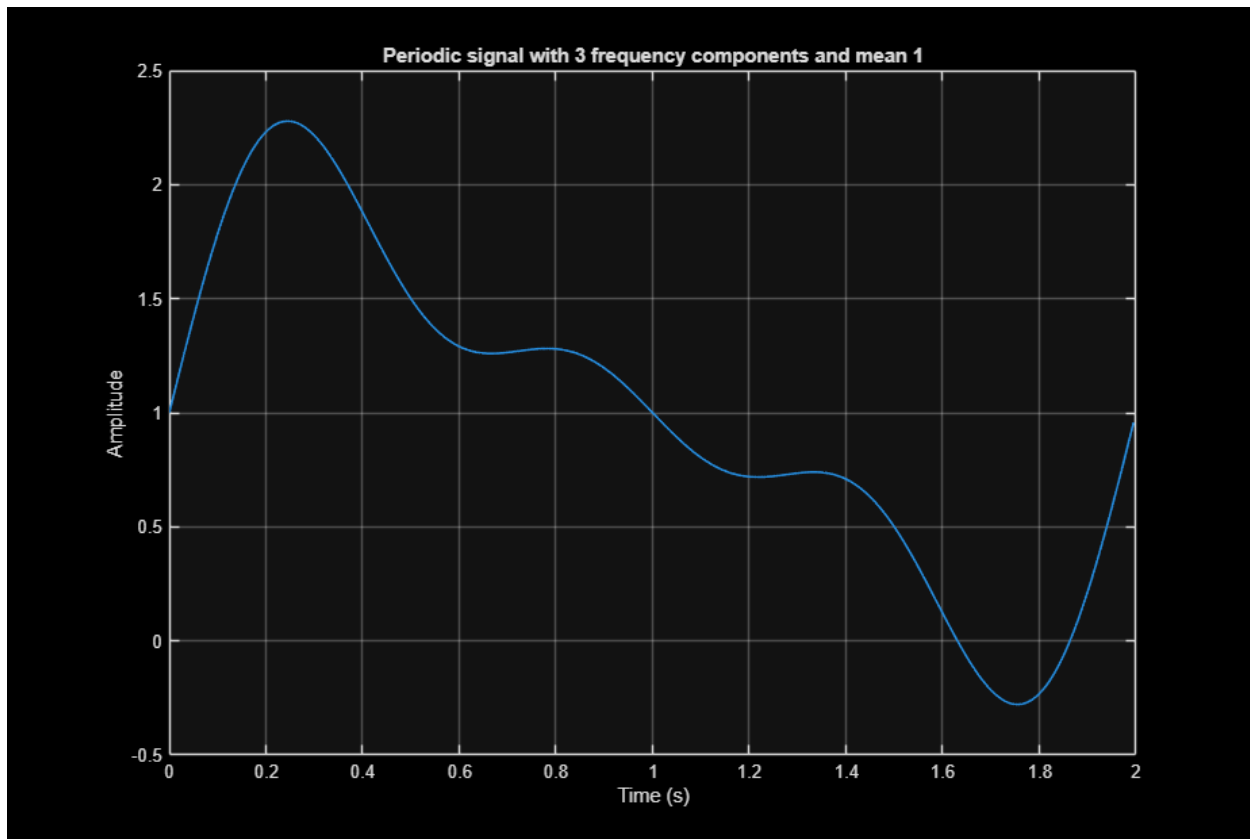
% Corresponding amplitudes (arbitrary but non-zero)
A = [0.8    0.5    0.3];    % amplitudes

% Optional phases - here we choose zero for simplicity
phi = [0 0 0];                % phases (radians)

% ----- Construct the signal -----
x = 1; % mean offset
for k = 1:length(f)
    x = x + A(k)*sin(2*pi*f(k)*t + phi(k));
end

% ----- Plot -----
figure;
plot(t, x, 'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Amplitude');
title('Periodic signal with 3 frequency components and mean 1');
grid on;
xlim([0 T]);    % display the full 0-2 s interval

```



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